



THOMAS COX

UNIVERSITY STUDENT

Brisbane, Australia

ABOUT ME

I am currently in my final year of university. Over the course of these years, I have become passionate about electrical engineering and physics whilst gaining academic experience. As an empathetic communicator, I aim to truly understand those around me as well as the topics I am invested in, striving to achieve the best result that I can through my efficient multitasking skills.

SKILLS

Technological Skills

- Programming: C, MATLAB, Arduino, Python
- Data Analysis: Signal processing, MATLAB simulations, control system Simulation
- Electronics: Circuit analysis, PCB design
- Proficient in LaTeX
- 3D modelling in Blender and Unity

Project Management

- Team collaboration
- Technical documentation

Soft Skills

- Time management - through university
- Critical thinking and problem-solving
- Adaptability - taking on different roles

Completed Projects

github.com/totomcox/portfolio

Publications

Recycled Thermocol Nanofibers Based Smart Triboelectric Nanogenerators for AI-Assisted Switching
<https://doi.org/10.1002/sml.202512532>

Triboelectric Nanogenerator – Creation and Characterisation

- Worked with postdoctoral advisors to design and build a triboelectric nanogenerator.
- Demonstrated uses including a digital switch and a highly accurate pedometer using 3D modelling software, Matlab, and Arduino

Bifunctional Organic Diodes – Light Emission and Detection

- In progress: fabricating and testing diodes for emission and sensing.
- Completing full electrical-optical testing to track performance and wear.

Other Major Projects

- "Simon Says" Game on a microprocessor
- PCB design for a dual-tone alarm
- DC Motor
- Design of Super-Heterodyne Receiver
- Movie Search Database

EDUCATION

Queensland University of Technology
Double Bachelors of Engineering and Physics
Expected to be completed Nov 2026

- Recently completed my undergraduate physics thesis in nanogenerators
- Halfway through my undergraduate engineering thesis in bifunctional organic diodes
- Worked with key academics, contributing to their research papers
- Currently in Final Year of Electrical Engineering (Honours) and Physics

Key Courses Completed and Related Skills

- **Engineering Sustainability & Professional Practice** — Design methods, sustainability, ethics, risk management
- **Foundation of Engineering Design** — Design principles, requirements analysis, systems engineering
- **Engineering Mechanics** — Statics, dynamics, mechanical system analysis
- **Microprocessors & Digital Systems** — Embedded systems, microcontrollers, digital logic
- **Foundations of Electrical Engineering** — Circuit analysis, AC/DC systems, op-amps
- **Signal Analysis** — Fourier/Laplace/Z-transforms, filtering, spectral analysis
- **Electronic Design** — PCB design, prototyping, analog/digital design
- **Telecommunications & RF** — Modulation, RF components, transmission lines
- **Control & Dynamic Systems** — PID control, modelling, MATLAB/Simulink
- **Power Electronics** — Converters, inverters, switching devices
- **Electromagnetics & Machines** — Maxwell's equations, electric machines
- **Electromagnetism** — EM waves, boundary conditions, transmission lines
- **Materials & Thermal Physics** — Materials, heat transfer, thermodynamics
- **Classical & Quantum Physics** — Classical mechanics, quantum fundamentals

Technical interests

- Triboelectrics
- Optoelectronics
- Embedded Systems
- Signal Processing
- Semiconductor physics
- Control Systems
- PCB Design

Technical Tools & Software

- MATLAB/Simulink
- LTspice
- Eagle (PCB)
- LabVIEW
- Python (NumPy, Matplotlib)
- Unity & Blender
- Git / GitHub
- Oscilloscopes, Function Generators, SMUs

CAREER HISTORY

"Dominos"

as multiple roles

Brisbane | August 2023 - October 2023

- Took on a range of different roles such as delivery and food prep where I improved my adaptability and communication skills

"Arthritis and Rheumatology Consultants"

as a Concierge

Minnesota | May 2020 - December 2020

- Conducted patient intake assessments, analyzing responses to guide individuals through the clinic efficiently, improving workflow.

"Noodles and Company"

as a Cook

Minnesota | June 2019 - January 2020

- Managed high-volume meal preparation, ensuring quality and consistency for 50-100 customers per hour in a fast-paced environment.

REFEREES

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HOW TO REACH ME

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